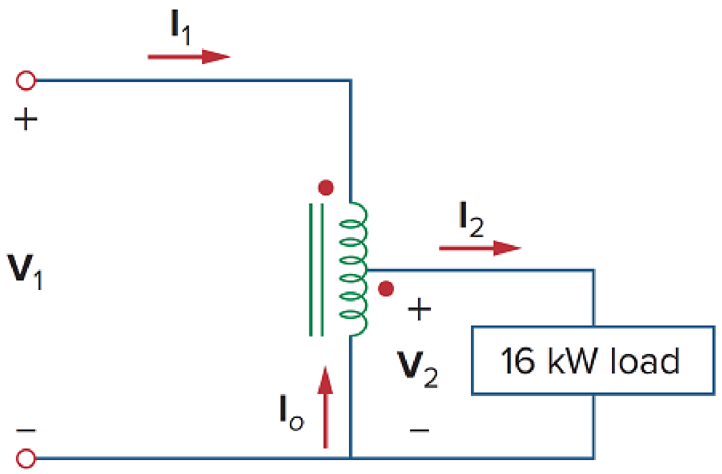
Chapter 13, Problem 11PP

Problem

In the autotransformer circuit of Fig. 13.45, find currents **I**1, **I**2, and **I***o*. Take **V**1 = 8 kV, **V**2 = 2 kV.

Figure 13.45



Step-by-step solution

Step 1 of 4

Refer to Figure 13.45 from the textbook.

Write the expression for the power delivered to the load, p_2.

$p_2 = V_2 I_2$ ...... (1)

Here,

V_2 is the secondary side voltage of the auto transformer,

I_2 is the secondary side current of the auto transformer.

Step 2 of 4

Substitute 2 kV for V_2 and 16 kW for p_2 in equation (1).

$$16\times10^3 = \left(2\times10^3\right) I_2$$

$$I_2 = \frac{16}{2}$$

$$I_2 = 8\text{ A}$$

Step 3 of 4

Since the power at both the ends of the transformer is same.

Thus,

$$p_1 = p_2$$
$$V_1 I_1 = V_2 I_2$$

Substitute 16 kV for V_1, 2 kV for V_2 and 8 A for I_2.

$$\left(8\times10^3\right) I_1 = \left(2\times10^3\right)(8)$$
$$I_1 = \frac{16\times10^3}{8\times10^3}$$
$$I_1 = 2\text{ A}$$

Step 4 of 4

Calculate the current, I_o.

Apply Kirchhoff's Current Law at the meeting point of I_1, I_2 and I_o.

$$I_o + I_1 - I_2 = 0$$
$$I_o = I_2 - I_1$$

Substitute 2 A for I_1 and 8 A for I_2.

$$I_o = 8 - 6$$
$$= 2\text{ A}$$

Therefore, the currents, I_1, I_2 and I_o are $\boxed{2\text{ A}}$, $\boxed{8\text{ A}}$ and $\boxed{6\text{ A}}$ respectively.